



CompMix: A Benchmark for Heterogeneous Question Answering

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ABSTRACT

Fact-centric question answering (QA) often requires access to multiple, heterogeneous, information sources. By jointly considering several sources like a knowledge base (KB), a text collection, and tables from the web, QA systems can enhance their answer coverage and confidence. However, existing QA benchmarks are mostly constructed with a single source of knowledge in mind. This limits capabilities of these benchmarks to fairly evaluate QA systems that can tap into more than one information repository. To bridge this gap, we release COMP MIX, a crowdsourced QA benchmark which naturally demands the integration of a mixture of input sources. COMP MIX has a total of 9,410 questions, and features several complex intents like joins and temporal conditions. Evaluation of a range of QA systems on COMP MIX highlights the need for further research on leveraging information from heterogeneous sources.

CCS CONCEPTS

• Information systems → Question answering.

KEYWORDS

Question Answering, Heterogeneous Sources, Benchmark

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1 INTRODUCTION

Motivation. There is a long line of research on fact-based question answering (QA), largely divided into three directions: (i) methods that use a large curated knowledge base (KB) such as Wikidata [32] as the source of answers (KB-QA) (e.g., [1, 2]), (ii) methods that operate on a text corpus (text-QA) (e.g., [15, 36]), and (iii) methods that tap into a collection web tables (table-QA) (e.g., [14, 16]). Each of these directions has its own benchmarks for testing and comparing QA systems [3, 4, 14, 18, 34].

However, using only a single information source limits the answer coverage of QA systems: the individual sources are not complete, and may fail to cover the knowledge required for answering a user question. Consider, as an example, the question:

Who was fouled before the first penalty in the 2022 FIFA final?

This kind of detailed information on a sports event is rarely covered in a KB or table, but can be found in news and discussions about the match. On the other hand, structured sources often include information that is not present in text. Tables often store match-specific details, and would contain, for instance, the answer to the following question:

Argentina's ball possession in the 2022 WC final?

For some questions, answers appear in multiple sources. Such answer redundancy can also be helpful for QA systems, and boost their confidence in predicted answers. For instance, consider:

In which stadium was the 2022 soccer world cup final played?

The answer to this question occurs in a Wikipedia infobox, text content, and Wikidata. It may even be necessary to join evidence from multiple sources for answering a more complex question:

Which team was behind by two goals but still won a FIFA final?

The list of FIFA finals and winners can be looked up in a KB, but the goal deficit information on the match timeline occurs only in text or statistics tables. These observations have triggered work on heterogeneous QA [22, 27, 29, 30, 33]: jointly harnessing multiple sources for answering factual questions [26].

Benchmark limitations. There are two ways of evaluating QA over combinations of KB, text and tables: (i) starting with an existing single-source benchmark of questions, extend or vary the underlying sources, for example by adding text articles to a KB and optionally weakening the KB content so as to show the benefits of the added source (e.g., [22, 27, 29, 30, 33]); or (ii) using one of the few benchmarks dedicated to heterogeneous QA [7, 20, 31, 35].

The first approach is unsatisfying, as it creates artificial situations regarding the sources. In the second line, existing benchmarks are dominated by simple questions that are not really challenging for the QA system. Also, there is little diversity in formulations and user intents, and the benchmarks tend to cover only narrow domains of topics [6–8, 19, 37].

A new benchmark. We make the case for a new benchmark that inherently requires the usage of a mixture of information sources, as a more natural testbed for evaluating heterogeneous QA systems. To this end, we release COMP MIX (Complete questions over a Mixture of sources), a crowdsourced QA benchmark with questions that require heterogeneous sources for answering (Wikidata KB, and Wikipedia text, tables and infoboxes). The dataset has 9,410 questions created by humans from five different domains: books, movies, music, TV series and soccer. The answers are provided both as text and grounded to the Wikidata KB, for consistent evaluation across QA systems with different output formats.



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Contributions. This paper presents our benchmark COMPmix, accompanied by an in-depth analysis. We investigate the effect of combining multiple sources on answer coverage and redundancy, and show that heterogeneous sources are truly required.

Finally, we evaluate multiple recent heterogeneous QA methods on COMPmix, and identify questions for which none of these systems gives correct answers. Interestingly, results show that a recent GPT model can answer only half of the questions for this realistic and challenging benchmark. The COMPmix benchmark is publicly available at <https://qa.mpi-inf.mpg.de/compmix>.

2 BENCHMARK DESIGN PRINCIPLES

Prior benchmarks. There are many datasets for KB-QA (like WebQuestions [3] or SimpleQuestions [4]), text-QA (like SQuAD [24] or NaturalQuestions [18]), and table-QA (like WikiTableQuestions [23] or NQ-Tables [14]). However, these benchmarks were created with the intention of having a specific source for answering, which contains almost all answers to the questions.

All existing benchmarks for heterogeneous QA suffer from one or more of the following issues: (i) their questions are not fully human-generated, and hence lack the diverse formulations of real users [20, 31, 35]; (ii) they span only two sources, like tables and text [6, 8, 37], or text and KBs [20, 35]; (iii) they are restricted to small or artificial KBs, orders of magnitude smaller than large curated KBs like Wikidata [20, 35]; (iv) they explore only one domain like finance [8, 37], geography [19], or e-commerce [28]; and (v) their questions are only in conversational form with implicit intent, unsuitable for evaluating stand-alone QA methods [10, 12, 21].

Design principles. To overcome these limitations, we devised a new benchmark for heterogeneous QA benchmarks, with the following desiderata and resulting design principles:

- (i) Questions should be created by humans.
- (ii) Questions should naturally require heterogeneous sources.
- (iii) Sources should not be artificially constructed or restricted.
- (iv) The benchmark should not be domain-specific, but cover a wide spectrum of topics.
- (v) Questions should be self-contained and complete.

Our new benchmark COMPmix satisfies these design principles: (i) it is crowdsourced; (ii) it spans four kinds of sources; (iii) it includes the full large-scale KB as one of the sources; (iv) it covers five different thematic domains; and (v) its questions are self-contained. Table 1 gives a high-level comparison of various benchmarks (including COMPmix).

3 BENCHMARK DESCRIPTION

We create COMPmix by collecting the *completed* (or *intent-explicit*) versions of the questions in the CONVMix [10] benchmark, which is a dataset for conversational QA over heterogeneous sources that has *incomplete* (or *intent-implicit*) questions. These completed questions are provided directly by crowdworkers on Amazon Mechanical Turk (AMT), i.e. are created by *humans*. The answers to the questions were derived from *four* sources: either the full Wikidata KB, or the text, tables or infoboxes from all of Wikipedia. Basic statistics for COMPmix can be found in Table 2.

COMPmix includes questions, their domains, and their corresponding answers. Answers are Wikidata identifiers (text labels are

Table 1: Comparing benchmarks for heterogeneous QA.

Dataset	KB	Text	Table	Info	OR	HQ	OD
HYBRIDQA [7]	✗	✓	✓	✗	✗	✓	✓
MULTIMODALQA [31]	✗	✓	✓	✗	✓	✗	✓
OTT-QA [6]	✗	✓	✓	✗	✓	✓	✓
MANYMODALQA [13]	✗	✓	✓	✗	✗	✓	✓
WIKIMOVIES [20]	✓	✓	✗	✗	✓	✗	✗
TAT-QA [37]	✗	✓	✓	✗	✗	✓	✗
FINQA [8]	✗	✓	✓	✗	✗	✓	✗
HETPQA [28]	✗	✓	✓	✓	✗	✓	✗
COMPmix (ours)	✓	✓	✓	✓	✓	✓	✓

OR: Open Retrieval; HQ: Human Questions; OD: Open Domain.

Table 2: Basic statistics for the COMPmix benchmark.

Domains	Books, Movies, Music, TV series, Soccer
Questions	9,410 (train: 4,966, dev: 1,680, test: 2,764)
Avg. question length	9.19 words (min=2, median=9, max=28)
Avg. no. of question entities	1.11 (min=1, median=1, max=4)
Avg. answer length (text)	2.17 words (min=1, median=2, max=21)
Avg. no. of answers	1.02 (min=1, median=1, max=6)
Entities covered	5,413 (long-tail: 2,511, with <50 KB-facts)

also provided), plaintext strings, or normalized dates. This enables consistent evaluation across extractive and generative answering models. Entity markup in questions is provided by crowdworkers. Answer sources are given, too: “KB”, “text”, “table”, or “infobox”.

Ethics. COMPmix collates completed questions from the CONVMix benchmark. For collecting CONVMix, human annotators from AMT asked factoid questions in a conversational setting. The COMPmix benchmark does not contain any personal or other critical data. All questions are provided anonymously.

4 BENCHMARK ANALYSIS

Answer coverage. One key desideratum of the benchmark is that heterogeneous sources are actually required for answering the questions inside. We thus analyzed the answer coverage of each information source, which is the number of questions that a source contains the answer for. In a good benchmark for heterogeneous QA, each source should have an answer coverage far less than 100%.

At the time of benchmark creation, Turkers provided the source they consulted for locating their answer. For computing coverage, we first consider these *source annotations by the crowdworkers*. However, this measurement only captures whether a specific information source has the desired information, without any implications concerning the other sources.

Therefore, we also conducted an *automatic analysis of the answer coverage* using a recall-oriented retriever that, given a question, tries to obtain as many relevant pieces of evidence as possible from all our sources. This retriever is implemented as in [10, 11], and first disambiguates KB-entities from the question (using CLOCQ [9], a recent system), and then retrieves KB-facts, text-sentences, table-records and infobox-entries with these disambiguated KB-entities. For each evidence, mentions of entities are linked to the KB. We measure this *automated answer coverage* as the number of questions for which the gold answer is among this set of mentioned entities in the pool of retrieved evidence. As with any large-scale automated analysis, this statistic is a noisy proxy, because the mere presence of

Table 3: Answer coverage across information sources.

Source(s)	Annotated	Automated
KB	0.308	0.807
Text	0.280	0.690
Tables	0.112	0.272
Infoboxes	0.299	0.545
KB+Text	0.588	0.853
KB+Tables	0.420	0.821
KB+Infoboxes	0.607	0.831
Text+Tables	0.393	0.702
Text+Infoboxes	0.580	0.734
Tables+Infoboxes	0.412	0.610
KB+Text+Tables	0.701	0.857
KB+Text+Infoboxes	0.888	0.861
KB+Tables+Infoboxes	0.720	0.841
Text+Tables+Infoboxes	0.692	0.743
All sources	1.000	0.865

Table 4: Answer redundancy across information sources.

Answer found in 1 source	0.157
Answer found in 2 sources	0.168
Answer found in 3 sources	0.341
Answer found in all sources	0.199

an answer does not necessarily mean that the surrounding evidence is question-relevant.

The results of both analyses are in Table 3. First, we see that the AMT annotators used the KB, text and infoboxes almost equally often to answer their questions (tables also consulted $\geq 10\%$ of times). This proves that **CompMix is not biased towards any specific underlying source**. Second, from the automated measurement, we learn that **adding an information source always improves the answer coverage**. Note that this is a natural expansion, as opposed to augmentation after artificial suppression of large parts of specific sources. Also, our recall-oriented retriever only provides a loose upper bound ($\approx 87\%$): the performance of an actual retriever that balances recall and precision would currently reach a lower number (cf. Sec. 5). Thus, our benchmark leaves substantial room for the development of smart *heterogeneous retrievers*. Overall, these measurements suggest that all four sources are naturally required for answering the questions in CompMix, and different sources complement each other nicely.

Answer redundancy. Answer redundancy creates scope to test a heterogeneous system’s ability to boost confidence in its prediction when multiple matches happen across sources. For each question, we thus measured the number of sources touched by the retrieved pieces of evidence that actually contain the gold answer. Results are in Table 4. What we can see from here is that for a substantial proportion of questions, the answer is located in two ($\approx 17\%$) or three ($\approx 34\%$), out of four, sources. A reasonable chunk even has redundancy across all sources ($\approx 20\%$). This shows that CompMix has ample answer redundancy to be exploited.

Anecdotal examples. Table 5 shows representative examples from the CompMix benchmark. The examples illustrate that our dataset has a wide range of questions in terms of both *syntactic structure* – from well-formulated fluent questions (4, 5) to ad hoc telegraphic queries (1, 3), as well as *semantic complexity* – from simple intents (1) to more complex ones requiring conjunction (2), temporal understanding (3, 5), or aggregations (4).

5 EVALUATION WITH COMPMIX

Metrics. We use standard QA metrics for evaluating models on CompMix: (i) *Precision at 1* ($P@1$), which is either 1 or 0 according as the top-ranked system answer is correct or not; (ii) *Mean reciprocal rank* (MRR), which is the reciprocal of the first rank at which a correct answer is located; and, (iii) *Hit at 5* ($\text{Hit}@5$), which is either 1 or 0 according as the first five system responses contains a gold answer or not. A system answer is considered correct if it exactly (case-insensitive) matches a Wikidata ID (if QA system returns IDs) or the accompanying plaintext string/entity label (if QA system returns simple text). Metrics are averaged over all questions.

Models. To better understand the state-of-the-art in heterogeneous QA, we evaluate several recent QA models that incorporate heterogeneous sources on CompMix. We also include GPT in our model suite, to verify if LLMs trained on colossal web corpora are already sufficient for this task. We compare the following models:

- **UniK-QA** [22] follows a retriever-reader pipeline, and verbalizes evidence from each source into text. DPR [17] retrieves relevant evidences from the verbalized text, and a Fusion-in-decoder (FiD) model [15] generates the answer. Due to unavailability of end-to-end source code, we approximate UniK-QA by replacing DPR with BM25 [25]. FiD generates strings, that are mapped to a ranked list of KB items, by following [11].
- **CONVINSE** [10] is a method for conversational QA over heterogeneous sources, but can also be applied to complete questions. It derives an intent-explicit structured representation for a question, and feeds this into a retriever-reader pipeline.
- **EXPLAIGNN** [11] is another method for heterogeneous QA that makes use of iterative graph neural networks for deriving the answer instead of a generative reader model like FiD.
- **GPT-3**. For GPT-3 [5] (text-davinci-003), we use the following prompt, which performed the best among different variants: “Please answer the following question by providing the crisp answer entity, date, year, or numeric number. Q: <question>”. The generated answer is compared with the gold answers and their KB-aliases to allow for potential synonymy. $P@1$ is 1 for exact matches, and 0 otherwise. GPT-3 generates only a single answer, and thus metrics for ranked lists are inapplicable.

6 LESSONS LEARNED

The results in Table 6 reveal two key take-aways:

- Even the best state-of-the-art methods merely reach ca. 45% $P@1$ on CompMix, showing substantial room for further advances. Much higher numbers have been reported for these methods on previous benchmarks with easier questions (e.g., UniK-QA reaches 80% accuracy on WebQuestionsSP), highlighting the difficulty of CompMix;
- The QA task is far from solved even for LLMs, with $P@1$ by GPT-3 being merely 50%. We attribute this to a large number of rare and emerging entities in our benchmark (see Table 2).

To put the aggregate numbers in perspective, we found that for 2,764 questions (81.9%), at least one of the methods *failed* to produce a correct answer. On the other hand, for 759 (27.5%) *none* of the methods (including GPT-3) could find the correct answer. Interesting cases from this bucket include “Who played as adult Pi Patel in *Life of Pi* movie?” and “Who were the twin brothers who played soccer

Table 5: Representative questions from COMPmix. Sources that can be used for answering these questions are in brackets.

Books	Movies	Music	TV series	Soccer
1. <i>Author of the book To Kill a Mockingbird?</i>	2. <i>Which lead actress appeared in both Terms of Endearment and The Evening Star?</i>	3. <i>Who replaced Ozzy Osbourne in Black Sabbath the first time?</i>	4. <i>How many episodes of The 100 did Jason Rothenberg write?</i>	5. <i>Where did the Uruguay national football team play their first recorded match?</i>
Harper Lee [KB, Text, Table, Info]	Shirley MacLaine [KB]	Ronnie James Dio [Text, Info]	16 [KB, Text, Table]	Paso del Molino [Text]

Table 6: Heterogeneous QA models on COMPmix (test set).

Method ↓ / Metric →	P@1	MRR	Hit@5
UNI-K-QA [22]	0.440	0.467	0.494
CONVINSE [10]	0.407	0.437	0.483
EXPLAIGNN [11]	0.442	0.518	0.617
GPT-3 [5] (text-davinci-003)	0.502	–	–

for Manchester United?”. These make a perfect case for merging multiple sources, as subtle cues like “adult Pi Patel” or “twin brothers” are likely mentioned in textual sources, while movie cast or club membership is more easily looked up via structured repositories.

Overall, unlike prior benchmarks, our new resource clearly brings out the challenging nature of heterogeneous QA, posing large potential for advancing QA systems.

7 CONCLUSION

We developed and released COMPmix, a benchmark for heterogeneous QA that inherently requires the usage of multiple sources, and thus poses a challenging test to QA systems. Answering questions in COMPmix requires systems to work consistently well for intents spread across five domains, and deal with a wide variety of human formulations, with some asking about rare entities. This new resource can hopefully facilitate progress towards more robust QA that can judiciously leverage complementary sources of information for high recall, as well as redundancy across sources to boost precision.

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